

Directional Derivatives, Chain Rule, and Optimization

Lecture 4 for Multivariable Calculus

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1 Lecture Scope and Goals

In this lecture, we build directly on Lecture 3. The main goals are:

- define directional derivatives and connect them to the gradient,
- apply chain rule in one-parameter and multi-parameter settings,
- interpret the Jacobian matrix as the best linear map,
- solve unconstrained optimization problems in two variables,
- solve constrained optimization problems using Lagrange multipliers.

Central Theme

Partial derivatives describe change along coordinate axes. Directional derivatives and the chain rule tell us how functions change along *any* motion.

2 Directional Derivatives and Gradient

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, point $a = (a, b)$, and direction vector $u = (u_1, u_2)$ with $\|u\| = 1$.

2.1 Definition

The directional derivative of f at a in direction u is

$$D_u f(a) = \lim_{t \rightarrow 0} \frac{f(a + tu) - f(a)}{t},$$

if the limit exists.

2.2 Key Formula (Differentiable Case)

If f is differentiable at a , then

$$D_u f(a) = \nabla f(a) \cdot u.$$

So the directional derivative is the projection of $\nabla f(a)$ onto direction u .

2.3 Geometric Meaning

- $\nabla f(a)$ points in the direction of steepest increase.
- The maximum directional derivative is $\|\nabla f(a)\|$, achieved at $u = \frac{\nabla f(a)}{\|\nabla f(a)\|}$.
- The minimum directional derivative is $-\|\nabla f(a)\|$.
- If $u \perp \nabla f(a)$, then $D_u f(a) = 0$ (instantaneously no change).

Example 1. Let

$$f(x, y) = x^2y + y^3.$$

Then

$$\nabla f = (2xy, x^2 + 3y^2).$$

At $(1, 1)$,

$$\nabla f(1, 1) = (2, 4).$$

For direction $u = (\frac{3}{5}, \frac{4}{5})$,

$$D_u f(1, 1) = \nabla f(1, 1) \cdot u = (2, 4) \cdot \left(\frac{3}{5}, \frac{4}{5}\right) = \frac{22}{5}.$$

Pitfall 1

The direction vector must be a *unit* vector in the formula $D_u f = \nabla f \cdot u$. If a non-unit vector v is used, normalize first: $u = \frac{v}{\|v\|}$.

3 Tangent Plane and Normal Line (Applied View)

For $z = f(x, y)$ differentiable at (a, b) :

$$z \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

This is the tangent plane approximation.

3.1 Normal Vector to Tangent Plane

For surface $F(x, y, z) = f(x, y) - z = 0$, a normal vector is

$$\nabla F(a, b, f(a, b)) = (f_x(a, b), f_y(a, b), -1).$$

Any nonzero scalar multiple is also normal, so both n and $-n$ are valid normals.

3.2 How to Choose the Normal *Direction*

A surface has two opposite normal directions at a smooth point. In applications we often need a consistent choice (an *orientation*).

1) Implicit surface $F(x, y, z) = c$. The gradient ∇F points toward increasing values of F . So:

- $+\nabla F$ and $-\nabla F$ are both normal directions,
- choose the sign using words in the problem: *outward*, *upward*, *inward*, etc.

2) Graph surface $z = f(x, y)$. For $F(x, y, z) = f(x, y) - z$, a normal is $(f_x, f_y, -1)$; the opposite normal is $(-f_x, -f_y, 1)$. The second one has positive z -component, so it is the *upward* normal.

3) Parametric surface $\mathbf{r}(u, v)$. Use

$$\mathbf{n} = \mathbf{r}_u \times \mathbf{r}_v.$$

Reversing parameter order flips direction:

$$\mathbf{r}_v \times \mathbf{r}_u = -(\mathbf{r}_u \times \mathbf{r}_v).$$

So orientation depends on parameter order.

Example (Sphere). For sphere $x^2 + y^2 + z^2 = R^2$, take

$$F(x, y, z) = x^2 + y^2 + z^2, \quad \nabla F = (2x, 2y, 2z).$$

At point $p = (x_0, y_0, z_0)$, $\nabla F(p)$ points away from the origin, so it is the *outward* normal direction. The inward normal is $-\nabla F(p)$.

Example 2. For $f(x, y) = x^2 + xy + y^2$ at $(1, 2)$:

$$f(1, 2) = 7, \quad f_x = 2x + y, \quad f_y = x + 2y,$$

so

$$f_x(1, 2) = 4, \quad f_y(1, 2) = 5.$$

Tangent plane:

$$z = 7 + 4(x - 1) + 5(y - 2).$$

A normal vector is $(4, 5, -1)$.

3.3 Example 2 via Level Curve (Mathematica View)

For

$$f(x, y) = x^2 + xy + y^2, \quad f(1, 2) = 7,$$

the level curve through $(1, 2)$ is

$$x^2 + xy + y^2 = 7.$$

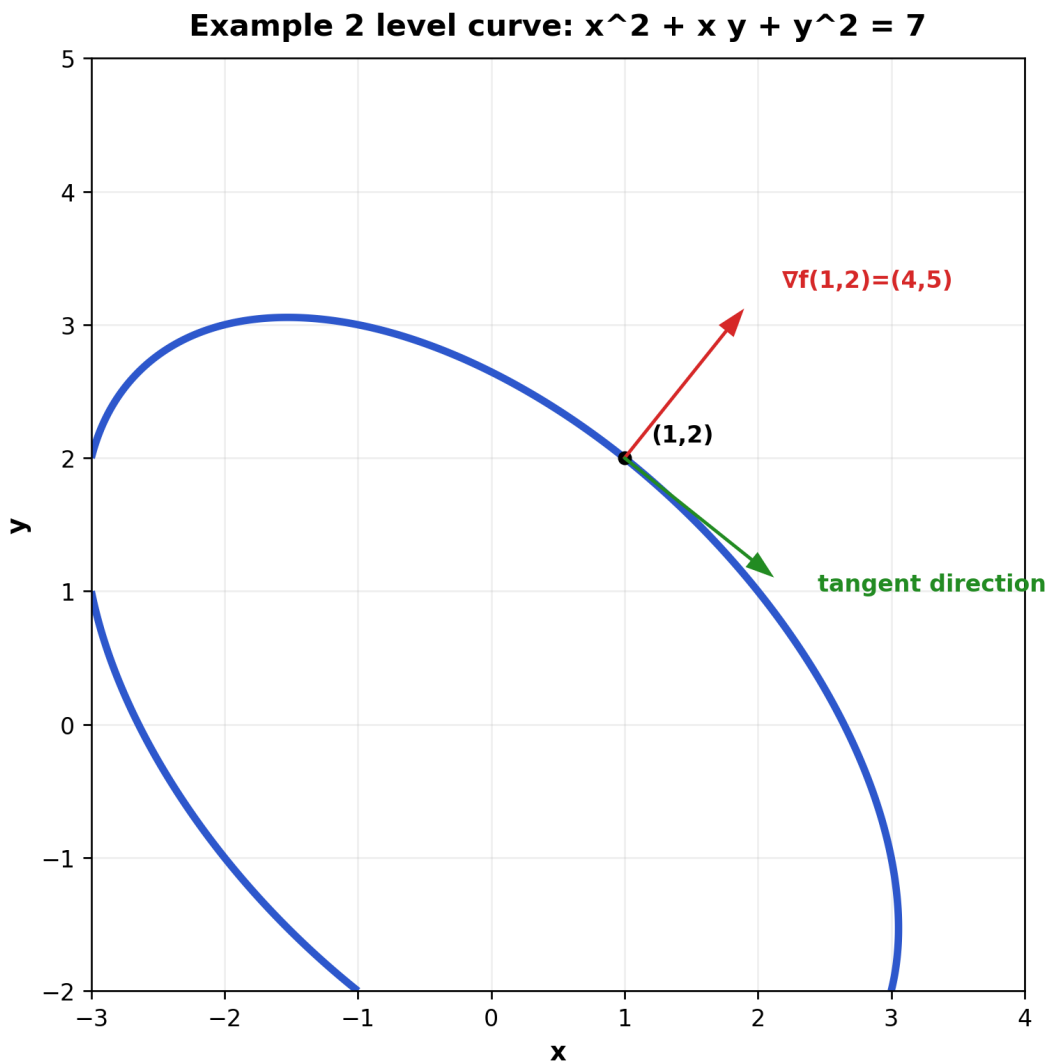


Figure 1: Level curve $f(x, y) = 7$ with the gradient vector and a tangent direction at $(1, 2)$.

At $(1, 2)$,

$$\nabla f(1, 2) = (f_x(1, 2), f_y(1, 2)) = (4, 5).$$

For any tangent vector t to the level curve, f is constant along the curve, so

$$\frac{d}{ds}f(\gamma(s)) = \nabla f \cdot \gamma'(s) = \nabla f \cdot t = 0.$$

Hence $\nabla f(1, 2)$ is normal to the level curve in the xy -plane. For the graph surface $F(x, y, z) = f(x, y) - z = 0$, this becomes the 3D normal

$$\nabla F(1, 2, 7) = (4, 5, -1),$$

which is exactly the normal used in the tangent-plane equation above.

3.4 Visualizing Normal Direction in 3D

Normal Direction on $z = x^2 + y^2$ at $(1,1,2)$

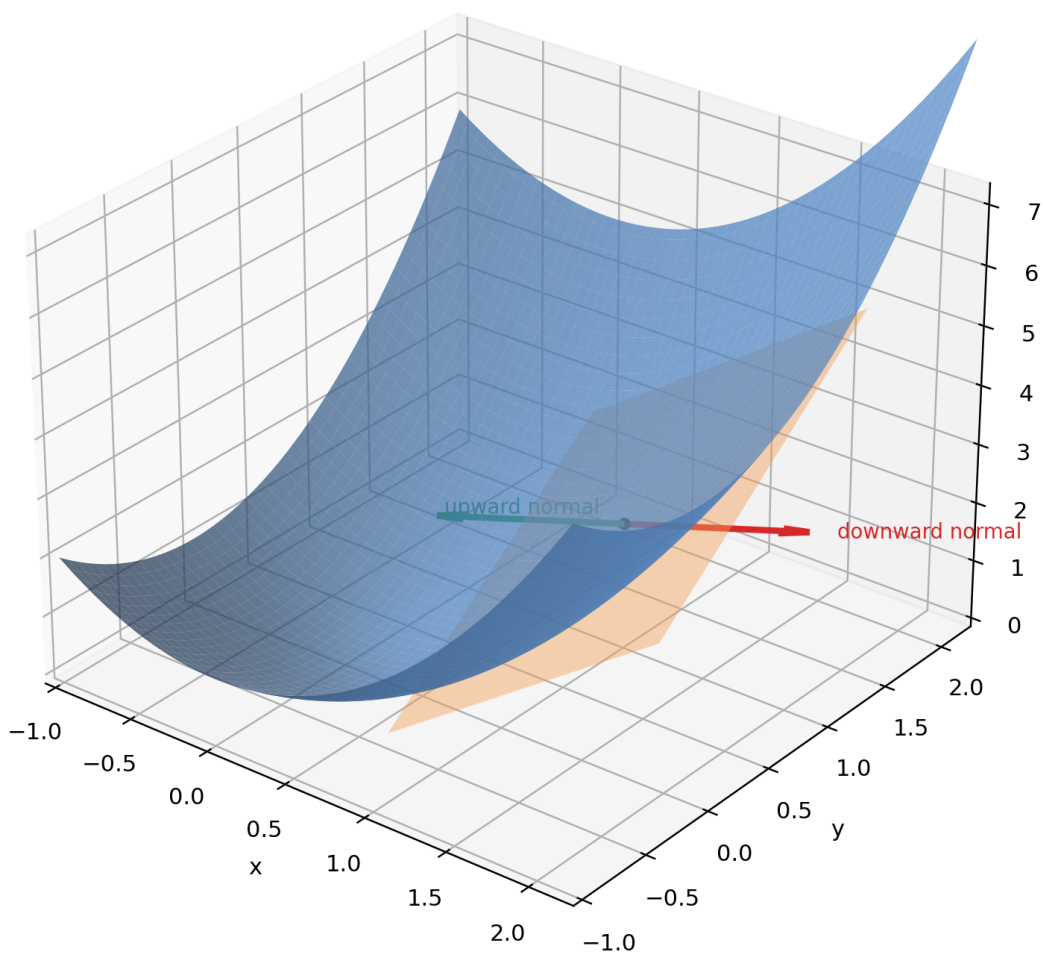


Figure 2: For graph surface $z = x^2 + y^2$, two opposite normal directions at the same point correspond to opposite choices of orientation.

Normal Direction on $x^2 + y^2 + z^2 = 4$

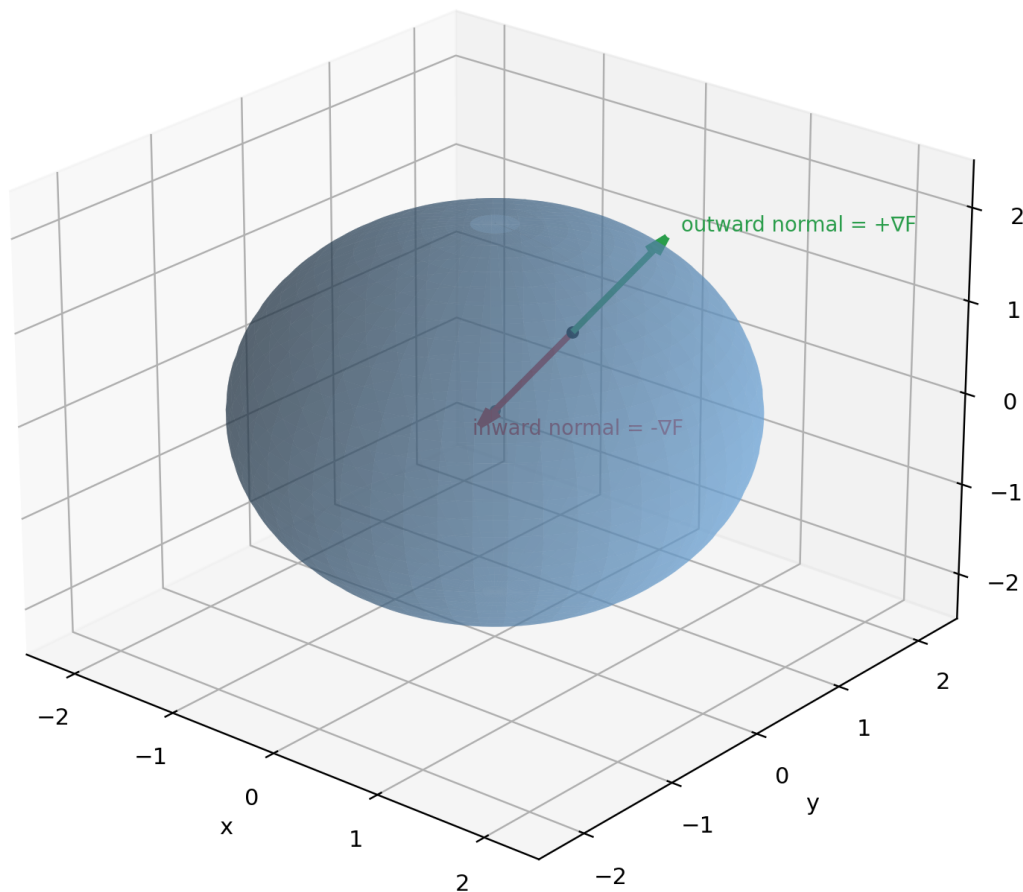


Figure 3: For sphere $x^2 + y^2 + z^2 = 4$, ∇F gives the outward normal direction and $-\nabla F$ gives the inward normal direction.

4 Chain Rule in Multivariable Calculus

4.1 Why Gradient Gives Normal Direction (Geometric Meaning)

Let a level surface be

$$F(x, y, z) = c.$$

Take any smooth curve $\mathbf{r}(t)$ lying on this surface. Then $F(\mathbf{r}(t)) = c$, so

$$\frac{d}{dt}F(\mathbf{r}(t)) = 0.$$

By the chain rule,

$$\frac{d}{dt}F(\mathbf{r}(t)) = \nabla F(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = 0.$$

Here $\mathbf{r}'(t)$ is a tangent vector to the surface, so ∇F is orthogonal to every tangent direction. Therefore ∇F is a normal vector of the level surface.

Geometric meaning.

- ∇F points in the direction of fastest increase of F .
- $\|\nabla F\|$ is the maximal rate of increase per unit distance.
- Along tangent directions, first-order change is zero, so tangent directions are perpendicular to ∇F .

4.2 Case A: One Parameter t

Let $x = x(t)$, $y = y(t)$ and $z = f(x, y)$. Then

$$\frac{dz}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt} = \nabla f \cdot (x'(t), y'(t)).$$

Example 3. Let $f(x, y) = x^2y$, with $x = t^2$, $y = e^t$.

$$\frac{d}{dt}f(x(t), y(t)) = f_x x' + f_y y'.$$

Compute:

$$f_x = 2xy, \quad f_y = x^2, \quad x' = 2t, \quad y' = e^t.$$

So

$$\frac{dz}{dt} = 2xy(2t) + x^2e^t = 4txy + x^2e^t.$$

Substitute $x = t^2$, $y = e^t$:

$$\frac{dz}{dt} = 4t(t^2)e^t + t^4e^t = (4t^3 + t^4)e^t.$$

4.3 Case B: Two Parameters (s, t)

If $x = x(s, t)$, $y = y(s, t)$ and $w = f(x, y)$, then

$$\frac{\partial w}{\partial s} = f_x \frac{\partial x}{\partial s} + f_y \frac{\partial y}{\partial s}, \quad \frac{\partial w}{\partial t} = f_x \frac{\partial x}{\partial t} + f_y \frac{\partial y}{\partial t}.$$

Example 4. Let $f(x, y) = e^{xy}$, $x = s + t$, $y = s - t$. Then

$$xy = (s + t)(s - t) = s^2 - t^2,$$

so

$$w(s, t) = e^{s^2 - t^2}.$$

Directly,

$$\frac{\partial w}{\partial s} = 2se^{s^2 - t^2}, \quad \frac{\partial w}{\partial t} = -2te^{s^2 - t^2}.$$

These match chain rule computations.

5 Jacobian Matrix and Linearization

For $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ with components $F = (F_1, \dots, F_m)$, the Jacobian at a is

$$J_F(a) = \left[\frac{\partial F_i}{\partial x_j}(a) \right]_{m \times n}.$$

5.1 Interpretation

If F is differentiable at a ,

$$F(a+h) = F(a) + J_F(a)h + o(\|h\|).$$

So $J_F(a)$ is the best linear approximation.

Example 5. Let

$$F(x, y) = (x^2 - y, xy).$$

Then

$$J_F(x, y) = \begin{bmatrix} 2x & -1 \\ y & x \end{bmatrix}.$$

At $(1, 2)$:

$$J_F(1, 2) = \begin{bmatrix} 2 & -1 \\ 2 & 1 \end{bmatrix}.$$

6 Unconstrained Optimization in Two Variables

6.1 Critical Points

For $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, a point (a, b) is critical if

$$\nabla f(a, b) = (0, 0)$$

or one/both first partials do not exist.

6.2 Second Derivative Test

Let

$$A = f_{xx}(a, b), \quad B = f_{xy}(a, b), \quad C = f_{yy}(a, b), \quad D = AC - B^2.$$

Then:

- If $D > 0$ and $A > 0$: local minimum.
- If $D > 0$ and $A < 0$: local maximum.
- If $D < 0$: saddle point.
- If $D = 0$: inconclusive.

Example 6.

$$f(x, y) = x^3 - 3x + y^2.$$

First derivatives:

$$f_x = 3x^2 - 3, \quad f_y = 2y.$$

Critical points: $x = \pm 1, y = 0$, so $(1, 0)$ and $(-1, 0)$. Second derivatives:

$$f_{xx} = 6x, \quad f_{xy} = 0, \quad f_{yy} = 2.$$

At $(1, 0)$:

$$D = (6)(2) - 0 = 12 > 0, \quad A = 6 > 0 \Rightarrow \text{local minimum.}$$

At $(-1, 0)$:

$$D = (-6)(2) - 0 = -12 < 0 \Rightarrow \text{saddle point.}$$

Pitfall 2

Second derivative test classifies only local behavior near critical points. It does not automatically give a global maximum/minimum on unbounded domains.

7 Constrained Optimization: Lagrange Multipliers

To optimize $f(x, y)$ subject to $g(x, y) = c$, solve

$$\nabla f(x, y) = \lambda \nabla g(x, y), \quad g(x, y) = c.$$

7.1 Why It Works (Geometric Intuition)

At an optimal point on a smooth constraint curve, the level curve of f is tangent to the constraint. Their normals are parallel, so gradients are parallel.

Example 7. Find max/min of

$$f(x, y) = xy$$

subject to

$$x^2 + y^2 = 1.$$

Set

$$\nabla f = (y, x), \quad \nabla g = (2x, 2y),$$

so

$$(y, x) = \lambda(2x, 2y).$$

System:

$$y = 2\lambda x, \quad x = 2\lambda y, \quad x^2 + y^2 = 1.$$

From first two equations, $x = \pm y$. On unit circle:

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right), \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right), \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right).$$

Function values:

$$f = \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}.$$

So max is $\frac{1}{2}$, min is $-\frac{1}{2}$.

8 Extra In-Class Example Bank (Board-Ready)

These are short examples designed for live teaching. Each can be done in about 3–6 minutes.

8.1 Example 8 (Directional Derivative with a Non-Unit Direction)

Let

$$f(x, y) = x^2 - xy + 2y^2, \quad a = (1, -1), \quad v = (2, 1).$$

Compute $D_u f(a)$ where $u = \frac{v}{\|v\|}$.

Solution.

$$\nabla f = (2x - y, -x + 4y), \quad \nabla f(1, -1) = (3, -5).$$

Normalize direction:

$$\|v\| = \sqrt{5}, \quad u = \left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right).$$

Hence

$$D_u f(1, -1) = \nabla f(1, -1) \cdot u = (3, -5) \cdot \left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right) = \frac{1}{\sqrt{5}}.$$

8.2 Example 9 (Fastest Increase and Fastest Decrease)

Let

$$f(x, y) = e^x \cos y.$$

At $(0, \pi/3)$, find:

- direction of fastest increase,
- maximal rate of increase,
- direction of fastest decrease.

Solution.

$$\nabla f = (e^x \cos y, -e^x \sin y).$$

At $(0, \pi/3)$:

$$\nabla f\left(0, \frac{\pi}{3}\right) = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right).$$

Its norm is 1. Therefore:

- fastest increase direction: $\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$,
- maximal rate: 1,
- fastest decrease direction: $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$.

8.3 Example 10 (Chain Rule Along a Curve)

Let

$$f(x, y) = \ln(1 + x^2 + y^2), \quad x(t) = t^2, \quad y(t) = \sin t.$$

Find $\frac{d}{dt}f(x(t), y(t))$.

Solution.

$$f_x = \frac{2x}{1 + x^2 + y^2}, \quad f_y = \frac{2y}{1 + x^2 + y^2}, \quad x' = 2t, \quad y' = \cos t.$$

So

$$\frac{d}{dt}f(x(t), y(t)) = f_x x' + f_y y' = \frac{2x(2t) + 2y \cos t}{1 + x^2 + y^2}.$$

Substitute $x = t^2$, $y = \sin t$:

$$\frac{d}{dt}f(x(t), y(t)) = \frac{4t^3 + 2 \sin t \cos t}{1 + t^4 + \sin^2 t} = \frac{4t^3 + \sin(2t)}{1 + t^4 + \sin^2 t}.$$

8.4 Example 11 (Unconstrained Optimization: Two Critical Points)

Classify critical points of

$$f(x, y) = x^2 + xy + y^2 - 4x.$$

Solution.

$$f_x = 2x + y - 4, \quad f_y = x + 2y.$$

Solve

$$2x + y - 4 = 0, \quad x + 2y = 0.$$

From $x = -2y$, substitute into first equation:

$$-4y + y - 4 = 0 \Rightarrow y = -\frac{4}{3}, \quad x = \frac{8}{3}.$$

So the only critical point is $(\frac{8}{3}, -\frac{4}{3})$. Second derivatives:

$$f_{xx} = 2, \quad f_{xy} = 1, \quad f_{yy} = 2.$$

Hence

$$D = f_{xx}f_{yy} - f_{xy}^2 = 4 - 1 = 3 > 0, \quad f_{xx} = 2 > 0.$$

Therefore $(\frac{8}{3}, -\frac{4}{3})$ is a strict local minimum (in fact global, since the quadratic form is positive definite).

8.5 Example 12 (Lagrange Multipliers on an Ellipse)

Find extrema of

$$f(x, y) = x + y$$

subject to

$$g(x, y) = \frac{x^2}{4} + y^2 = 1.$$

Solution.

$$\nabla f = (1, 1), \quad \nabla g = \left(\frac{x}{2}, 2y\right).$$

Set

$$(1, 1) = \lambda \left(\frac{x}{2}, 2y\right).$$

So

$$1 = \lambda \frac{x}{2}, \quad 1 = 2\lambda y \Rightarrow x = \frac{2}{\lambda}, \quad y = \frac{1}{2\lambda}.$$

Substitute into constraint:

$$\frac{1}{4} \left(\frac{2}{\lambda}\right)^2 + \left(\frac{1}{2\lambda}\right)^2 = 1 \Rightarrow \frac{1}{\lambda^2} + \frac{1}{4\lambda^2} = 1 \Rightarrow \frac{5}{4\lambda^2} = 1 \Rightarrow \lambda = \pm \frac{\sqrt{5}}{2}.$$

Thus candidate points are

$$\left(\frac{4}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right), \quad \left(-\frac{4}{\sqrt{5}}, -\frac{1}{\sqrt{5}}\right).$$

Function values:

$$f_{\max} = \frac{5}{\sqrt{5}} = \sqrt{5}, \quad f_{\min} = -\sqrt{5}.$$

9 Mini Workflow for Students

Use this checklist for Lecture 4 problems:

1. If asked for rate of change in direction, normalize direction vector first.
2. For surface normals, first compute a valid normal, then pick sign to match orientation words (*upward/outward/inward*).
3. Compute gradient, then use $D_u f = \nabla f \cdot u$.
4. For composite variables, write dependence graph and apply chain rule carefully.
5. For extrema without constraints: find critical points and use second derivative test.
6. For constraints: solve $\nabla f = \lambda \nabla g$ together with the constraint equation.

10 Practice Set (In-Class or Homework)

1. Let $f(x, y) = x^2 + 2xy - y^2$. Compute $D_u f(1, -1)$ for $u = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$.
2. Let $z = x^2 + y^2$, where $x = t \cos t$, $y = t \sin t$. Find $\frac{dz}{dt}$.
3. Find and classify critical points of $f(x, y) = x^2 + y^2 - 4x + 6y$.
4. Use Lagrange multipliers to find extrema of $f(x, y) = x + 2y$ on $x^2 + y^2 = 5$.
5. Let $F(x, y) = (x^2 + y, e^{xy})$. Compute $J_F(0, 1)$ and linear approximation near $(0, 1)$.

Additional Exercises: Multivariable Chain Rule

1. Let

$$z = f(x, y) = x^2y + \sin(xy), \quad x = t^2, \quad y = e^{-t}.$$

Compute $\frac{dz}{dt}$.

2. Let

$$w = f(x, y) = \ln(1 + x^2 + y^2), \quad x = s + t, \quad y = s^2 - t.$$

Compute $\frac{\partial w}{\partial s}$ and $\frac{\partial w}{\partial t}$.

3. Let

$$u = f(x, y, z) = xye^z, \quad x = r \cos \theta, \quad y = r \sin \theta, \quad z = r^2.$$

Compute $\frac{\partial u}{\partial r}$ and $\frac{\partial u}{\partial \theta}$.

4. Let

$$F(x, y) = (x^2 + y, xy^2), \quad G(u, v) = u^2 + 3v.$$

Define $H = G \circ F$. Compute $\nabla H(1, 2)$ using Jacobians (do not expand first).

5. Let

$$T(s, t) = (s^2 - t, e^{st}), \quad q(x, y) = \frac{x}{1 + y^2}.$$

Set $Q(s, t) = q(T(s, t))$. Compute Q_s and Q_t at $(1, 0)$.

6. Let

$$f(x, y) = x^3 - xy + y^2, \quad \gamma(t) = (\cos t, \sin(2t)).$$

Compute $\frac{d}{dt}f(\gamma(t))$ at $t = 0$ in two ways:

(i) chain rule via $\nabla f \cdot \gamma'(t)$, (ii) direct substitution.

Verify the two answers agree.

7. Let

$$w = f(x, y, z) = \sqrt{x^2 + y^2 + z^2}, \quad x = uv, \quad y = u - v, \quad z = u^2 + v^2.$$

Compute $\frac{\partial w}{\partial u}$ and $\frac{\partial w}{\partial v}$.

8. Let

$$f(x, y) = \arctan\left(\frac{y}{x}\right), \quad x = t, \quad y = t^2.$$

Find $\frac{d}{dt}f(x(t), y(t))$ for $t > 0$, and simplify to a one-variable formula.

Solution Sketches: Multivariable Chain Rule

1.

$$f_x = 2xy + y \cos(xy), \quad f_y = x^2 + x \cos(xy), \quad x' = 2t, \quad y' = -e^{-t}.$$

Hence

$$\frac{dz}{dt} = f_x x' + f_y y'$$

with $x = t^2$, $y = e^{-t}$:

$$\frac{dz}{dt} = \left(2t^2 e^{-t} + e^{-t} \cos(t^2 e^{-t}) \right) (2t) - \left(t^4 + t^2 \cos(t^2 e^{-t}) \right) e^{-t}.$$

2.

$$f_x = \frac{2x}{1+x^2+y^2}, \quad f_y = \frac{2y}{1+x^2+y^2}, \\ x_s = 1, \quad x_t = 1, \quad y_s = 2s, \quad y_t = -1.$$

So

$$w_s = f_x x_s + f_y y_s = \frac{2x + 4sy}{1+x^2+y^2}, \\ w_t = f_x x_t + f_y y_t = \frac{2x - 2y}{1+x^2+y^2},$$

then substitute $x = s + t$, $y = s^2 - t$.

3.

$$f_x = ye^z, \quad f_y = xe^z, \quad f_z = xye^z.$$

With $x = r \cos \theta$, $y = r \sin \theta$, $z = r^2$:

$$x_r = \cos \theta, \quad y_r = \sin \theta, \quad z_r = 2r, \\ x_\theta = -r \sin \theta, \quad y_\theta = r \cos \theta, \quad z_\theta = 0.$$

Thus

$$u_r = f_x x_r + f_y y_r + f_z z_r = e^{r^2} (2r \sin \theta \cos \theta + 2r^3 \sin \theta \cos \theta) \\ = 2re^{r^2} (1 + r^2) \sin \theta \cos \theta, \\ u_\theta = f_x x_\theta + f_y y_\theta = r^2 e^{r^2} (\cos^2 \theta - \sin^2 \theta) = r^2 e^{r^2} \cos(2\theta).$$

4. Jacobians:

$$J_F(x, y) = \begin{bmatrix} 2x & 1 \\ y^2 & 2xy \end{bmatrix}, \quad J_G(u, v) = \begin{bmatrix} 2u & 3 \end{bmatrix}.$$

At $(1, 2)$:

$$F(1, 2) = (3, 4), \quad J_G(F(1, 2)) = \begin{bmatrix} 6 & 3 \end{bmatrix},$$

$$J_F(1, 2) = \begin{bmatrix} 2 & 1 \\ 4 & 4 \end{bmatrix}.$$

So

$$\nabla H(1, 2) = J_H(1, 2) = J_G(F(1, 2)) J_F(1, 2) = \begin{bmatrix} 24 & 18 \end{bmatrix}.$$

5. Let $x = s^2 - t$, $y = e^{st}$, so $Q = q(x, y) = \frac{x}{1 + y^2}$.

$$q_x = \frac{1}{1 + y^2}, \quad q_y = -\frac{2xy}{(1 + y^2)^2}.$$

Also

$$x_s = 2s, \quad x_t = -1, \quad y_s = te^{st}, \quad y_t = se^{st}.$$

At $(s, t) = (1, 0)$:

$$x = 1, \quad y = 1, \quad q_x = \frac{1}{2}, \quad q_y = -\frac{1}{2}, \quad x_s = 2, \quad x_t = -1, \quad y_s = 0, \quad y_t = 1.$$

Hence

$$Q_s = q_x x_s + q_y y_s = 1, \quad Q_t = q_x x_t + q_y y_t = -1.$$

6. Chain-rule method:

$$\nabla f = (3x^2 - y, -x + 2y), \quad \gamma'(t) = (-\sin t, 2 \cos 2t).$$

At $t = 0$: $\gamma(0) = (1, 0)$, $\nabla f(1, 0) = (3, -1)$, $\gamma'(0) = (0, 2)$.

$$\left. \frac{d}{dt} f(\gamma(t)) \right|_{t=0} = \nabla f(\gamma(0)) \cdot \gamma'(0) = (3, -1) \cdot (0, 2) = -2.$$

Direct method:

$$f(\gamma(t)) = \cos^3 t - \cos t \sin 2t + \sin^2 2t,$$

differentiate and evaluate at $t = 0$; result is also -2 .

7.

$$f_x = \frac{x}{w}, \quad f_y = \frac{y}{w}, \quad f_z = \frac{z}{w}, \quad w = \sqrt{x^2 + y^2 + z^2}.$$

And

$$x_u = v, \quad y_u = 1, \quad z_u = 2u, \quad x_v = u, \quad y_v = -1, \quad z_v = 2v.$$

Therefore

$$w_u = f_x x_u + f_y y_u + f_z z_u = \frac{xv + y + 2uz}{w},$$

$$w_v = f_x x_v + f_y y_v + f_z z_v = \frac{xu - y + 2vz}{w},$$

with $x = uv$, $y = u - v$, $z = u^2 + v^2$.

8. For $t > 0$, $x = t$, $y = t^2$:

$$f_x = -\frac{y}{x^2 + y^2}, \quad f_y = \frac{x}{x^2 + y^2}, \quad x' = 1, \quad y' = 2t.$$

So

$$\frac{d}{dt} f(x(t), y(t)) = f_x x' + f_y y' = -\frac{t^2}{t^2 + t^4} + \frac{t}{t^2 + t^4} (2t) = \frac{1}{1 + t^2}.$$

Takeaway

Lecture 4 extends derivative ideas from local linearization to motion, direction, and optimization. The gradient unifies these topics: it controls directional change, powers chain rule structure, and identifies extrema with or without constraints.

11 Selected Exercises from the Attached Book (English Translation)

The following problems are selected and translated from your attached PDF (*Section 6.6: Directional Derivative and Gradient*), with notation adjusted for this lecture.

1. Let

$$f(x, y) = x^2 - xy + y^2.$$

At

$$P_0 = (2 + \sqrt{2}, 1 + 2\sqrt{2}),$$

compute the directional derivative along a unit direction making angle θ with the positive x -axis. Then determine:

- for which θ the directional derivative is maximal,
- for which θ it is minimal,
- for which θ it equals 0.

2. Let

$$f(x, y) = x^3y + 3xy^2 + 2.$$

Find the directional derivative at $P_0 = (3, 1)$ in the direction from P_0 to $P = (6, 5)$.

3. Let

$$f(x, y) = \ln(x - y).$$

Find the directional derivative at $(1, 2)$ along the tangent direction to the parabola

$$y = 2x^2$$

at that point.

4. Let

$$u(x, y, z) = xy + yz + z^2.$$

Find the directional derivative at

$$P_0 = (2, 1, 3)$$

in the direction that makes equal angles with the positive x -, y -, and z -axes.

5. Let

$$f(x, y) = x^2 + 2xy + y^2.$$

Compute the gradient at $(1, 2)$.

6. Let

$$f(x, y) = \arctan\left(\frac{y}{x}\right).$$

Compute $\nabla f(x_0, y_0)$ at a general point (x_0, y_0) with $x_0 \neq 0$, and then compute the directional derivative at (x_0, y_0) along the direction of the vector (x_0, y_0) .

7. Let f be a differentiable homogeneous function of degree n in three variables, i.e.

$$f(tx, ty, tz) = t^n f(x, y, z) \quad \text{for all } t \in \mathbb{R}.$$

Prove Euler's identity:

$$xf_x + yf_y + zf_z = nf(x, y, z).$$

8. Let $f(x, y)$ be differentiable on the punctured plane and satisfy

$$yf_x - xf_y = 0.$$

Show that f depends only on the radius $r = \sqrt{x^2 + y^2}$; equivalently, there exists G such that

$$f(r \cos \theta, r \sin \theta) = G(r).$$

Solution Sketches

1.

$$\nabla f = (2x - y, -x + 2y), \quad \nabla f(P_0) = (3, 3\sqrt{2}).$$

For $u = (\cos \theta, \sin \theta)$:

$$D_\theta f(P_0) = \nabla f(P_0) \cdot u = 3 \cos \theta + 3\sqrt{2} \sin \theta = 3\sqrt{3} \cos(\theta - \arctan \sqrt{2}).$$

So max is $3\sqrt{3}$, min is $-3\sqrt{3}$, and $D_\theta f = 0$ when

$$\theta = -\arctan\left(\frac{1}{\sqrt{2}}\right) + k\pi.$$

2.

$$\nabla f = (3x^2y + 3y^2, x^3 + 6xy), \quad \nabla f(3, 1) = (30, 45).$$

Direction from $(3, 1)$ to $(6, 5)$ is $(3, 4)$, unit vector $u = (3/5, 4/5)$.

$$D_u f(3, 1) = \nabla f(3, 1) \cdot u = 30 \cdot \frac{3}{5} + 45 \cdot \frac{4}{5} = 54.$$

3. As written, $f(x, y) = \ln(x - y)$ is not real-valued at $(1, 2)$, so this is likely a typo in the source. Using the standard corrected form $f(x, y) = \ln(y - x)$:

$$\nabla f = \left(-\frac{1}{y-x}, \frac{1}{y-x}\right), \quad \nabla f(1, 2) = (-1, 1).$$

For $y = 2x^2$, tangent slope at $x = 1$ is $y' = 4$, so a tangent direction is $(1, 4)$, with unit vector $(1, 4)/\sqrt{17}$.

$$D_u f(1, 2) = (-1, 1) \cdot \frac{(1, 4)}{\sqrt{17}} = \frac{3}{\sqrt{17}}.$$

4.

$$\nabla u = (y, x + z, y + 2z), \quad \nabla u(2, 1, 3) = (1, 5, 7).$$

Equal angles with positive axes $\Rightarrow$ unit direction

$$u = \frac{1}{\sqrt{3}}(1, 1, 1).$$

Hence

$$D_u u(2, 1, 3) = \nabla u(2, 1, 3) \cdot u = \frac{1 + 5 + 7}{\sqrt{3}} = \frac{13}{\sqrt{3}}.$$

5.

$$f(x, y) = x^2 + 2xy + y^2 = (x + y)^2, \quad \nabla f = (2x + 2y, 2x + 2y).$$

At $(1, 2)$:

$$\nabla f(1, 2) = (6, 6).$$

6.

$$f(x, y) = \arctan\left(\frac{y}{x}\right), \quad \nabla f(x_0, y_0) = \left(-\frac{y_0}{x_0^2 + y_0^2}, \frac{x_0}{x_0^2 + y_0^2}\right).$$

Direction of (x_0, y_0) uses

$$u = \frac{(x_0, y_0)}{\sqrt{x_0^2 + y_0^2}}.$$

Then

$$D_u f(x_0, y_0) = \nabla f(x_0, y_0) \cdot u = 0.$$

7. From

$$f(tx, ty, tz) = t^n f(x, y, z),$$

differentiate both sides with respect to t :

$$xf_x(tx, ty, tz) + yf_y(tx, ty, tz) + zf_z(tx, ty, tz) = nt^{n-1}f(x, y, z).$$

Set $t = 1$:

$$xf_x + yf_y + zf_z = nf(x, y, z).$$

8. In polar coordinates,

$$f_\theta = -yf_x + xf_y.$$

Given $yf_x - xf_y = 0$, we get $f_\theta = 0$, so f does not depend on θ . Therefore

$$f(r, \theta) = G(r),$$

i.e. $f(x, y)$ depends only on $r = \sqrt{x^2 + y^2}$.

12 Simple Derivation: Laplacian in Polar Coordinates (2D)

Let

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r = \sqrt{x^2 + y^2},$$

and let

$$u(x, y) = U(r, \theta).$$

We want a chain-rule derivation of

$$\Delta u = u_{xx} + u_{yy}$$

in terms of (r, θ) .

1) First derivatives by chain rule

$$\begin{aligned} r_x &= \frac{x}{r} = \cos \theta, & r_y &= \frac{y}{r} = \sin \theta, \\ \theta_x &= -\frac{y}{r^2} = -\frac{\sin \theta}{r}, & \theta_y &= \frac{x}{r^2} = \frac{\cos \theta}{r}. \end{aligned}$$

So

$$\begin{aligned} u_x &= U_r r_x + U_\theta \theta_x = U_r \cos \theta - \frac{1}{r} U_\theta \sin \theta, \\ u_y &= U_r r_y + U_\theta \theta_y = U_r \sin \theta + \frac{1}{r} U_\theta \cos \theta. \end{aligned}$$

2) Use directional decomposition of ∂_x, ∂_y

From the same chain rule,

$$\partial_x = \cos \theta \partial_r - \frac{\sin \theta}{r} \partial_\theta, \quad \partial_y = \sin \theta \partial_r + \frac{\cos \theta}{r} \partial_\theta.$$

Apply once more:

$$u_{xx} + u_{yy} = \left(\cos \theta \partial_r - \frac{\sin \theta}{r} \partial_\theta \right)^2 U + \left(\sin \theta \partial_r + \frac{\cos \theta}{r} \partial_\theta \right)^2 U.$$

When expanded, mixed terms cancel, and the remaining terms simplify to

$$\Delta u = U_{rr} + \frac{1}{r} U_r + \frac{1}{r^2} U_{\theta\theta}.$$

3) Final polar Laplace equation

Therefore, in polar coordinates:

$$\Delta u = u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta}$$

and Laplace's equation becomes

$$u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} = 0.$$

4) Common radial-only case

If $u = u(r)$ (no θ dependence), then $u_{\theta\theta} = 0$, so

$$u_{rr} + \frac{1}{r} u_r = 0.$$

13 Project-Style Problem: Multivariable Calculus in MLE (Least Squares Regression)

Background. Suppose we observe data

$$\{(x_i, y_i)\}_{i=1}^n, \quad x_i \in \mathbb{R}^p, \quad y_i \in \mathbb{R},$$

and use the linear model

$$y_i = x_i^\top \beta + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2) \text{ i.i.d.}$$

Let $X \in \mathbb{R}^{n \times p}$ be the design matrix, $y \in \mathbb{R}^n$ the response vector, and $\beta \in \mathbb{R}^p$ unknown parameters.

Project Tasks

1. Write the log-likelihood $\ell(\beta, \sigma^2)$ of the model.
2. For fixed σ^2 , show that maximizing ℓ in β is equivalent to minimizing

$$J(\beta) = \frac{1}{2} \|X\beta - y\|_2^2.$$

3. Compute $\nabla_\beta J(\beta)$ using multivariable chain rule.
4. Compute the Hessian $\nabla_\beta^2 J(\beta)$ and explain why J is convex.
5. Derive the normal equation

$$X^\top X \hat{\beta} = X^\top y.$$

When $X^\top X$ is invertible, give the closed-form solution.

6. Derive the MLE of variance:

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \|y - X\hat{\beta}\|_2^2.$$

7. Implement gradient descent

$$\beta^{(k+1)} = \beta^{(k)} - \eta \nabla_\beta J(\beta^{(k)}),$$

and compare with the closed-form $\hat{\beta}$ on a small dataset.

8. Briefly discuss: how does regularization (ridge: $J_\lambda(\beta) = \frac{1}{2} \|X\beta - y\|^2 + \frac{\lambda}{2} \|\beta\|^2$) change the gradient and solution?

Deliverables

- One-page derivation summary (equations and key calculus steps).
- Numerical experiment (table/plot of convergence of gradient descent).
- Short interpretation: parameter meaning and residual analysis.